

Industrial Series 500kVA UPS

Critical Power Infrastructure Specifications

System Overview

The Industrial Series 500kVA is a high-performance, industrial-grade, three-phase, double-conversion online UPS. It is engineered specifically for the rigorous demands of heavy-duty industrial environments and critical data center infrastructure, providing seamless power protection with zero tolerance for downtime.

Input / Output Electrical Parameters

Parameter	Specification Details
Input Voltage	380/400/415V AC (3-Phase + N + PE)
Input Frequency	50/60Hz
Operating Tolerance	-25% to +20%
Output Voltage	380/400/415V AC (Configurable)
Voltage Regulation	±1%
Total Harmonic Distortion (THDv)	< 2% (for 100% linear load)
System Efficiency	Up to 96.5% in high-efficiency double-conversion mode

Operating Environment & Temperature Limits

The system is designed to maintain peak performance across a variety of industrial conditions, ensuring reliability for critical equipment.

- **Operating Range:** 0°C to +40°C at full rated load.
- **Thermal Derating:** Required for operation in extended environments up to +50°C.
- **Humidity:** 0–95% relative humidity (non-condensing).

Battery Management & Energy Storage

Advanced storage compatibility allows for integration with modern multi-tier supply chain equipment.

- **Configuration:** Modular, hot-swappable battery racks for simplified maintenance and scalability.
- **Chemistry Compatibility:** Full support for VRLA (Valve Regulated Lead Acid) and LiFePO4 (Lithium Iron Phosphate) chemistries.
- **Intelligent BMS:** Integrated Battery Management System providing:
 - Real-time health and performance monitoring.
 - Automated periodic discharge testing.
 - Predictive failure alerts and state-of-charge tracking.

Safety & Compliance Standards

The 500kVA system adheres to international safety and quality standards to ensure compliance with modern EPC project delivery requirements.

- **UPS Safety:** IEC/EN 62040-1
- **EMC/EMI Compliance:** IEC/EN 62040-2
- **Ingress Protection:** IP20 Rating
- **Certifications:** RoHS and CE Certified